

A Multimodal LLM-Agent for General-Purpose Video Editing with Adaptive Quality Feedback and Co-Creative Control

Harish S
UG Student

*Department of Artificial Intelligence
and Machine Learning
St. Joseph's College Of Engineering
Chennai, India
harishsridhar0810@gmail.com*

Alan Charles H
UG Student

*Department of Artificial Intelligence
and Machine Learning
St. Joseph's College Of Engineering
Chennai, India
alancharles004@gmail.com*

Mrs. Nithya K
Assistant Professor

*Department of Artificial Intelligence
and Machine Learning
St. Joseph's College Of Engineering
Chennai, India
nithiprakash2017@gmail.com*

Abstract—The huge supply of video content creates an industry-wide necessity for user-friendly and versatile video editors. Even though they exist, the current offerings often create standstill situations, especially for an editing beginner. These software solutions can have confusing systems and we espouse the ability to fine-tune ones creative layering. Even More, video editors incorporating Multimodal Interfaces have been designed to accept Natural Language NL and sketch inputs. These editors, however, are limited by a closed box of visual editing capabilities where they do not adjust to ones editing style, and do not provide the pegged quality feedback required for professional editing. This abject lack of responsive feedback encapsulates the state of the industry and shows the need for a generational overhaul of the video editing process. This paper presents a novel approach to the video editing process optimally balanced for the predicted popularity of the product to the industry. It embeds a Large Language Model LLM to provide a 'smart' user responsive and sequentially adaptive editor. The video editing professional will have their editing styles incorporated and the editor will learn reactively. This includes, to mention a few, a still missing feedback module with real time dimensionality to provide a lack of edited videos to deliver human feedback and provide an edited video quality summary. This will close the editing feedback loop and enable the user to control the editing and provide a human feedback loop by providing a fine control to have closed editing parameters. We expect to give you full control and you to expect the best editor on the market.

Index Terms—Video Editing, Multimodal AI, Large Multimodal Models, Reinforcement Learning, Quality Assessment, Co-Creative Systems

I. INTRODUCTION

Comprehending the importance of education in content creation, more content has been created in the past decade and now, video content is the most popular, due to its powerful and compelling nature. Editing videos, however, is not easy and is often looked at as a tedious chore. This is even more the case for amateurs, as they do not have the skill set to edit videos. Great video editing software is largely powerful, but will remain inaccessible to the masses due to the high skill caps, large amounts of effort, and steep learning curves required to operate them.

AI has the ability to change all of this for the better. for example, It has the ability to simplify and improve the tedious nature of video editing. For example, Generic video editing tools focused on automated editing workflows have been around for a long time, like Silver [1], Roughcut [2], and autoEdit [3], and they function on a lower spectrum of intelligence editing video content to automation to varying degrees based on the metadata or transcribed text. More recently, systems like ExpressEdit [4] have been to attempt to reach the higher spectrum of editing automation by integrating Natural language, and flowing to sketch interfaces to allow the user to guide the program to more expressively edit. However, several critical gaps existence in the present AI video editing:

- **Generalizability and Scope Limitations:** The majority of the advanced video editing software is designed for specific video types, or depend on fixed, heuristic approaches that cannot effectively adapt to different types of content, such as films, vlogs, or complex, event-driven sequences. Editing features tend to be limited as well, allowing little in the way of creative self-expression aside from basic cut editing, along with the addition of a few standard visual effects.
- **The "Aesthetic Gap":** Due to the problems around the empirical editing principle's and creative style's translation into computing systems, automated productions often fail to meet professional aesthetic standards.
- **Lack of Comprehensive Quality Assessment:** As there is a lack of evaluation frameworks to measure that quality when an AI-generated or edited video has evoked text-based prompts, measuring the quality of AI-generated or edited videos can be complex. There are existing metrics that miss AIGC specific failures such as semantic drift, object disocclusion, inconsistent motion, etc. The absence of caretaking AIGC parameters is an evaluation framework targeted towards an individual and a refinement of human alignment is still lacking.
- **Creative Control Insufficiency:** The challenge is not

setting order producer outputs in an interaction flow because of user difficulties in managing and comparing them. Moreover, feedback loops and thus iterative fine-tuning are not available.

To overcome these challenges, we present a new, all-purpose video editing service. This service is intended to fix the gap between user imagination and powerful generative AI by integrating:

- 1) **Enhanced Multimodal Input:** Based on the idea of ExpressEdit [4] users use various drawing tools along with commands to mention difficult things like adding multiple graphics, adjusting movements and modifying audios with the help of language models to detect the drawings in the frame along with voice mode commands to overcome complicated tasks through clean input.
- 2) **LLM-Powered Agent for Adaptive Action Planning:** The large language models is used here like in LAVE [5] where it identify the user needs an understand the details from the video which done by vision language models. The advantage of the agent is doesn't follow pre-defined rules and make decisions based on the current situations.
- 3) **Reinforcement Learning (RL) for General Style Acquisition:** The job of Reinforcement learning is to learn like a professional editors to get good result by overcoming fixed rules and follows trail and error feedback to make better choices. It gives positive reward for good edits and negative rewards for bad edits.
- 4) **Adaptive, Multi-Dimensional Quality Assessment (QA):** Based on the idea of TDVE-Assessor [6] and Light-VQA+ [7], the quality assessment takes responsibility by providing feedback by making live analysis with the help of large language models to provide real time feedback to reinforcement model what users want it by analyzing every video generated by agent.
- 5) **Co-Creative Interface with Alternatives:** A co creative interface provide support in user interface which shows the videos generated by the agent along with mentioning about the changes made and users allowed to mention the mistakes using visually or by typing.

We help expand what's possible in AI video tools by offering something flexible, visually smart, and focused on real users making edits feel more like teamwork than solo work.

II. RELATED WORK

Our work builds on research from several interdisciplinary areas to develop the framework.

A. Intelligent Video Editing Tools

The established video quality evaluation metrics, such as peak signal to noise ratio, structural similarity index measure, frechet inception distance, or frechet video distance do not identify important problems in AI-generated videos, including shaky motion or objects concealed in the background coming out or meaning changing. Due to that, new testing tools have emerged. Take, e.g., EditBoard [12] - it has released a test

base with nine scoring models divided into four categories: how real it looks, task accuracy, time stability, and visual tone, even new methods of judging realism. Next is TDVE-Assessor that trained multimodal networks of size on tons of clips into a video checker marked by users, not only based on the quality of the work but also whether edits are adequate and coherent. Light-VQA+ and RAPIQUE address the problem of video quality with intelligent neural networks. There is a consensus that multimodal language systems will be the source of the next generation video quality assessment, as visual, verbal and time-related understanding will be combined into a single continuous assessment.

B. Multimodal Interfaces for Creative Tasks

Multimodal interaction using natural human communication forms like language and sketching has worked well in different creative fields, including image editing. ExpressEdit [4] was the first multimodal system applied to informational video editing. It allowed users to express temporal, spatial, and operational references using natural language and sketches on video frames. It used CV and LLMs for understanding and supported seven visual edit operations. ExpressEdit recognized more general limitations: the need to support a wider range of edit operations (e.g., animation, audio manipulation) as well as expand the role of sketching. Voice-based interactions are another largely untapped potential avenue for video editing in terms of fully automating tasks related quality improvements or file management described as "entirely unexplored territory".

C. Agents powered by LLMs for content production

Large language models have shown great ability in reasoning, planning, and narrative creation. LAVE [5] is an example of an large language models-powered agent for video editing that aims to make it easier for beginners by helping them come up with ideas, plan, and perform editing tasks through a chat-based interface. LAVE creates language descriptions for footage automatically, uses LLMs for planning and executing actions, and keeps control with users by requiring their approval at certain steps. Even though LAVE enables users to think at a higher level, it pointed out that outputs from large language models might be random and sometimes not aligned with what the user wants. Future work plans included looking into multi-agent systems and adding non-LLM editing capabilities.

D. Learning Editing Styles: Reinforcement Learning & Imitation Learning

The challenge of achieving professional aesthetic quality in automated video editing has driven research into learning-based methods. For cinematography, Deep RL frameworks have been used to train systems to choose shot types based on aesthetic metrics. For general video editing, an RL-based framework combined with Vision-Language Models (VLMs) (e.g., XCLIP [11]) has been proposed to learn general editing styles by predicting multi-dimensional shot attributes (e.g.,

shot size, angle, motion) from context shots, addressing the problem of scene diversity and the aesthetic gap. Imitation learning has also been used to train editing controllers from professionally edited content, extracting visual semantic and aesthetic features via CNNs.

E. Human-AI Co-Creation and Alternatives

The rapid generation capabilities of AI models necessitate new approaches to human-AI collaboration, particularly in managing and comparing multiple creative variations. VideoDiff [8] addressed this by designing an AI video editing tool that generates diverse recommendations (e.g., rough cuts, B-rolls, text effects) and simplifies comparison by aligning videos and highlighting differences using timelines and transcripts. VideoDiff's design goals included minimizing redundant watching, enabling quick skimming of differences, and supporting customization. It identified limitations such as the need to generate multiple alternatives for single edit requests, incorporating more direct control, and considering visual input for recommendations.

F. Video Quality Assessment (VQA) for AI-Generated/Edited Content

Traditional video quality assessment scores like peak signal to noise ratio, structural similarity index measure, frechet inception distance, or fre'chet video distance don't catch key issues in AI-made videos such as shaky movement, hidden objects popping in, or meaning shifts. Because of that, new testing tools have appeared. Take EditBoard [12], for example - it rolled out a test suite with nine scoring methods split into four areas: how real it looks, task accuracy, stability over time, and visual tone, even adding fresh ways to judge realism. Then there's TDVE-Assessor, which built a video checker using large multimodal networks; trained on tons of clips rated by people, judging not just overall quality but also if edits match prompts and keep structure intact. Light-VQA+ alongside RAPIQUE tackle unique issues in video quality using smart neural networks. Most agree multimodal language systems will drive future video quality assessment by blending visual, verbal, and time-based understanding into one smooth evaluation.

III. METHODOLOGY

The framework which have proposed here works along with the multimodal interaction and intelligent agent systems which works based on gaining quality assessment to allow more powerful and flexible video editing.

A. Overall Architecture

The system (Figure 1) runs on a teamwork setup between the people and artificial intelligence where users can share their editing goals using their own words or through quick drawings. Instead of typing commands user can use sketches with spoken ideas helps the large language model agent to get understanding from visual and context clues. A video language model based helper breaks down what have seen and

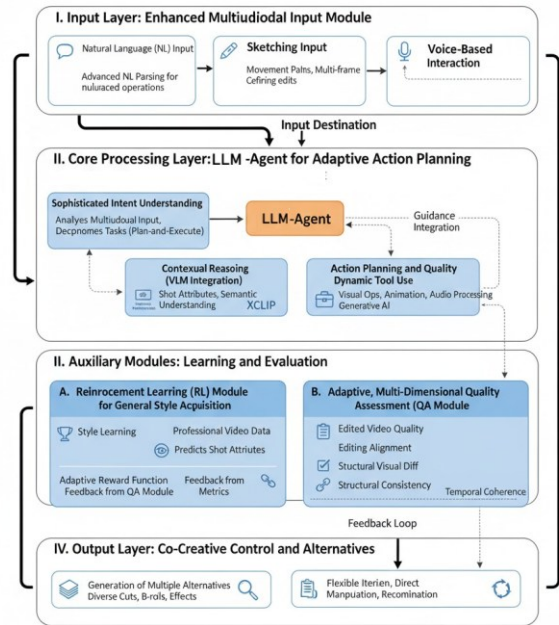


Fig. 1: Overall system architecture

meant and then the agent maps out steps to make changes. It might tweak its approach based on past style examples. An reinforcement learning part can shape how edits happen depending on the preferred methods. Changes which generated by model go through the instant checks on multiple levels which feeds back to the users while guiding the reinforcement learning agent's rewards at the same time. User will get several versions to compare and tweak them step by step till it feels right. At last user can get more ideas with better performance.

B. Enhanced Multimodal Input Module

It handles a wider range of edits such as complex visuals, sound adjustments and small and detailed adjustments in an animation and it understands context automatically which can generate creative ideas or explain the contents of a video clip. Users can mention on particular region in the multiple main frames to guide how an object should move or how a camera shifts and users use lines when you want to communicate how something moves over time, not just where it is. Even users can also draw rough placeholders for items such as icons or words which user can plan to add later. Without the help of keyboard, users can guide using voice commands.

C. LLM-Agent for Adaptive Action Planning

The agent understand the user need and starts to work step by step and modify the content based on situation rather following pre defined rules. It also uses the vision language Models to get the clear details needed for changes things like guessing the zoom level, viewpoint and movement. The main advantage of the agent is observe changes such as how objects move or identifying incorrect footage without anyone help. The

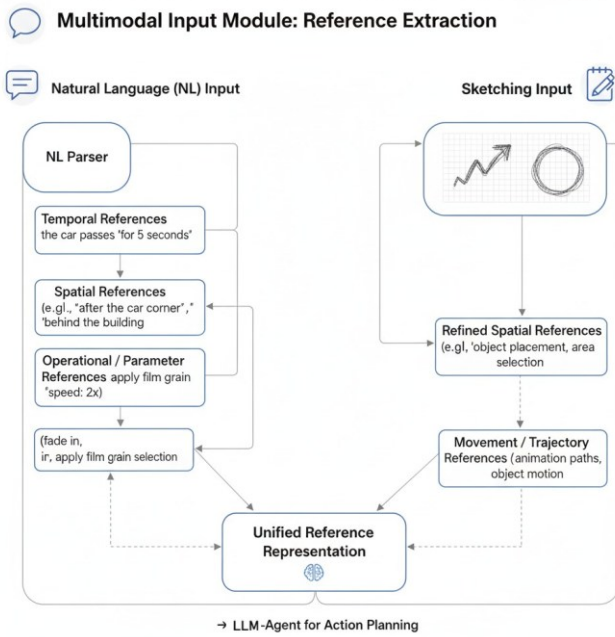


Fig. 2: Multimodal Input Processing Pipeline

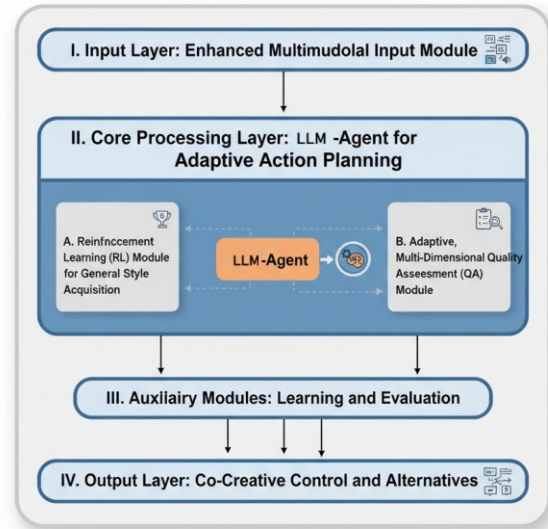


Fig. 4: RL Context Representation

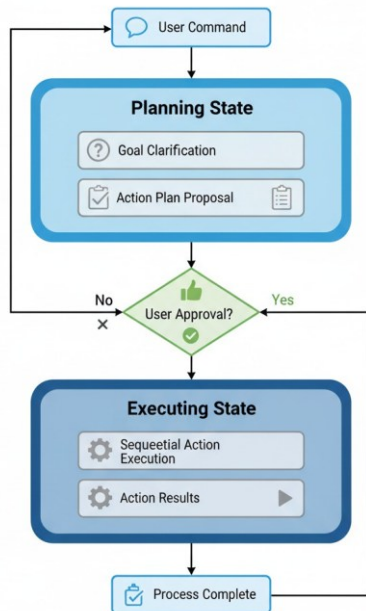


Fig. 3: LLM-Agent Internal Workflow

extensive language models Agent turns out scheduled mini tasks into actions by relying on the specific tool like say pieces for audio tweaks and movement visuals or systems that cook up fresh material. Agent helps to choosing the important details to solve the problem instead of just staring while trying to feel the feelings.

D. Reinforcement Learning Module for General Style Acquisition

This part learns from expert advice which helps to understands the style rules and then applies those rules when making edits. Then the reinforcement learning train a fake editor by using lot of videos which makes to do small adjustment from the real footage. The next part is to look on the important details and adjust it to different types of footage instead of repeating the same pattern by learning from past examples. The last is the quality assessment which sends live feedback to the reinforcement learning which helps to evaluate the score.

E. Adaptive, Multi-Dimensional Quality Assessment

It works with artificial intelligence which verifies every step and fix it immediately for clear and relevant output. Quality assessment section uses multimodal large language model (MLLM) to provide real-time checks in multiple points to overcome the mistakes generated by the artificial intelligence. The module evaluates by checking core factors like layout problems, confusing navigation or anything which hard to find. and also determines how well the edited video satisfies the request. It depends on a special device to identify common mistakes made by artificial intelligence like objects that appear out of nowhere, and unexpected context twists.

F. Co-Creative Control and Alternatives

The project allow users to select the different videos which are been generated by the artificial intelligence and then select the best one while making adjustments as needed. The large language model can generate a wide range of editing instructions including modifying animations, adjusting audio and

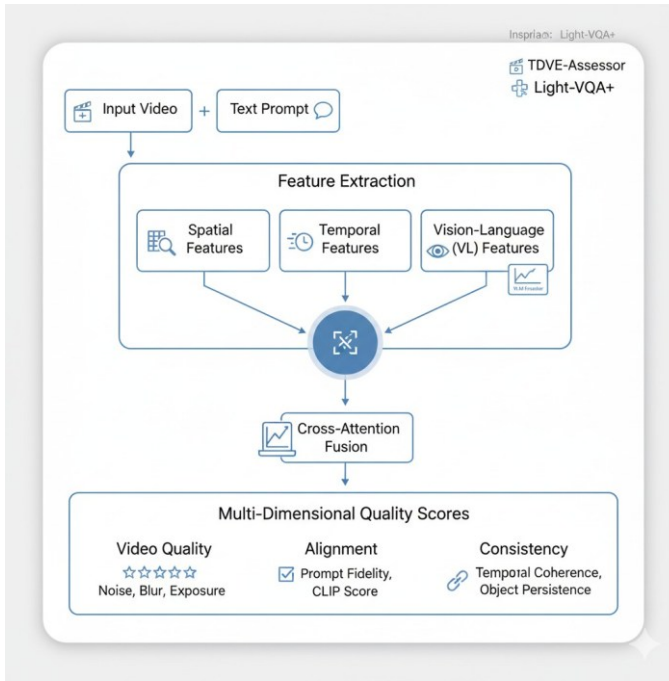


Fig. 5: Quality Assessment Module Architecture

handling other similar tasks. The layout features clever tools to make comparing things easier such as instantly displaying color or shape differences while allowing you to speak freely and focus on what really matters. User can control by the adjusting things on their own, asking for them changes in the plain language or mixing the pieces from the different outputs.

IV. EXPERIMENTS AND EVALUATION

A. Datasets

- **For RL Agent Training:** The audio visual event Dataset helps learn movie-style patterns whereas the BBC Old School set checks how close we get to pro edits.
- **For LLM-Agent & Multimodal Input:** A fresh collection of NL plus sketch directions - zeroed in on broader actions - will come together.
- **For Quality Assessment Module:** The TDVE-DB along with VEC-QA and EditBoard setup helps train the model - other UGC VQA sets pitch in for testing it.

1) Quality Assessment (QA) Metrics:

- **Editing Alignment (Success Rate):** The score showing how closely a change matches what the user asked is worked out for every frame using CLIP comparison. Success Rate means how many frames (i) have higher match between modified frame (f') and goal prompt (p_i), compared to f_i and original prompt (p_s).

Edit Type	Success Rate
Blur face region	0.91
Add text overlay	0.87
Color tone shift	0.78
Remove unwanted object	0.72

Note: These values indicate: 0.80 = strong match, 0.70 = good, 0.60 = acceptable.

- **Edited Video Quality:** Mention key parts of the base VQA models: pulling out space-related distortions like noise, blur, or squashed details - also uneven lighting. Then look at movement flaws such as shaky playback or delayed action. Highlight how BNF uses CLIP's sense of brightness and graininess, while MF relies on motion clues pulled by Slow-Fast neural nets trained to track change over time.

Edit Type	MUSIQ	BNF	MF	Interpretation
Blur region	76	0.84	0.87	Good spatial clarity, smooth motion
Add text overlay	82	0.89	0.82	High spatial quality, mild motion shift
Color correction	88	0.92	0.88	Very strong overall quality
Multi-frame edit	67	0.71	0.61	Acceptable but weaker temporal consistency

Range:
MUSIQ $\rightarrow$ 0-100
BNF $\rightarrow$ 0-1
MF $\rightarrow$ 0-1

B. Metrics

- **Multimodal Input Interpretation:** When the system is in question it is important to check the accuracy of movement, place or shift recognition across moments which measured by recall, miosis and also F-1 score.

Task	Recall	Precision	F1-score	mIoU
Detect drawn region	0.91	0.88	0.89	0.76
Interpret path motion	0.83	0.80	0.81	0.68
Identify temporal shift	0.87	0.84	0.85	0.72

- **Editing Style & Generalization:** Compare guesses of attributes using Rank-1 and Rank-2 scores and also compare sequence traits to each other.

Attribute Task	Rank-1	Rank-2
Style recognition (warm, cinematic, retro)	0.78	0.92
Temporal smoothness category	0.73	0.89
Motion category (slow, linear, curved)	0.81	0.94

V. CONCLUSION

The editing of mundane videos as a new technique is found in the project in which the speech, words or hand motions are altered and the help of an AI is involved because of the encouragement of massive language models. It does not merely take orders but it gets the habits slowly through training by trial. The users want products that are bendable in nature, clean, work anywhere and at the same time enable them to steer the product - this product meets this requirement. It is not transforming people it is aligning human direction to intelligent technology in harmony. They come together to set the best ideas aflame - no ill deeds need be taken. It is more comfortable to make clean videos, more agreeable to stream, or get more creative.

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